

**BEFORE YOU READ — ANSWER FROM MEMORY**

1. Name the **three** components of the achalasia triad.
2. Which esophageal cancer arises from Barrett, and in **which third** of the esophagus?
3. What single feature separates a **Mallory-Weiss tear** from **Boerhaave syndrome**?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 Obstructive & vascular disease

Achalasia — definition	Motility disorder from absence of the myenteric plexus in the body of the esophagus → increased LES tone and functional obstruction.
The triad	Increased LES tone + incomplete LES relaxation + esophageal aperistalsis.
Primary	Idiopathic — failure of distal esophageal inhibitory neurons .
Secondary	Chagas disease (<i>Trypanosoma cruzi</i>) destroys the myenteric plexus.
Achalasia-like (pseudo)	Diabetic autonomic neuropathy · infiltration by malignancy, amyloidosis or sarcoidosis · lesions of the dorsal motor nuclei (polio, surgical ablation).

Esophageal varices

What	Dilated veins of the submucosal venous plexus of the distal esophagus , prone to rupture and haemorrhage.
Cause	Cirrhosis and portal hypertension in >90% , mostly alcohol-induced. Also hepatic schistosomiasis.
Mechanism	Upper gastric and lower esophageal veins link to the portal system through gastroesophageal anastomoses; in portal hypertension these become prominent, dilated and varicose.
The number	Varices >5 mm may rupture → life-threatening bleeding. Overlying mucosa can be intact; ulcerated and necrotic once ruptured.



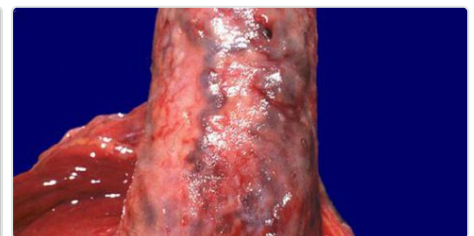
Achalasia: barium study and specimen



Barium swallow: serpiginous filling defects (varices)



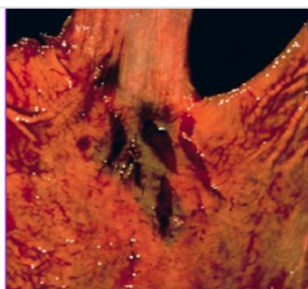
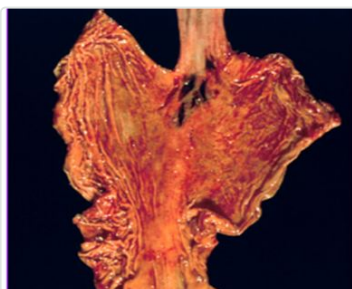
Endoscopy: large varices



Gross: dilated submucosal veins

2 Lacerations — the depth is the discriminator

	Mallory-Weiss tear	Boerhaave syndrome
Depth	Mucosal tear	Transmural tear and rupture
Site	Longitudinal, near the GE junction ; mm to several cm long	Distal esophagus
Trigger	Severe retching and vomiting , often acute alcohol intoxication; failure of reflex relaxation of GE musculature before the contractile wave	Similar factors, more severe
Presents with	GI bleeding, hematemesis	Severe chest pain, tachypnea, shock
Consequence	—	Severe mediastinitis ; usually fatal



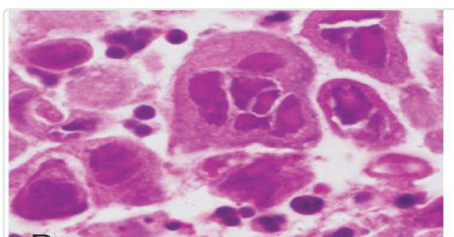
Mallory-Weiss: longitudinal mucosal tears



Boerhaave: transmural rupture

3 Esophagitis — one finding separates each

Causes & types	Reflux · chemical (alcohol, erosive acids, alkali) · medication (pill-induced, chemotherapy) · infectious · eosinophilic · radiation · desquamative skin disease (bullous pemphigoid, epidermolysis bullosa) · Crohn disease · graft-versus-host disease.	
Type	Who	The identifying finding
Herpes (HSV)	Debilitated / immunosuppressed	Shallow vesicles and ulcers that may coalesce; multinucleate squamous cells with herpesvirus nuclear inclusions
Candida	Debilitated / immunosuppressed	Adherent gray-white pseudomembranes of matted fungal hyphae and inflammatory cells
Eosinophilic	Allergic: other atopic diseases or peripheral eosinophilia	Transverse ridges resembling the trachea; large numbers of eosinophils in superficial epithelium, far from the GEJ
Reflux (GERD)	Adults >40 (also infants and children)	Eosinophils, then neutrophils infiltrate the epithelium; basal zone hyperplasia and elongated lamina propria papillae



Herpes: multinucleate cells, inclusions



Candida: pseudomembranes



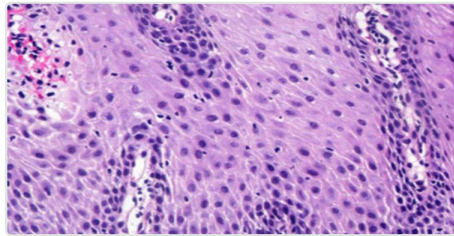
Eosinophilic: tracheal-like rings

Eosinophils appear in both reflux and eosinophilic esophagitis. The discriminator is **number and location**: scattered in reflux, numerous and far from the GEJ in eosinophilic esophagitis.

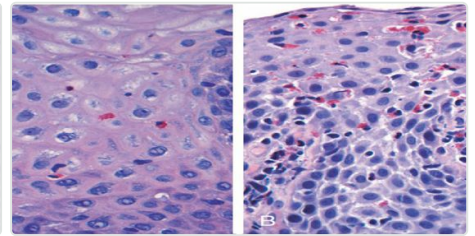
Reflux esophagitis	Inflammation of the lower esophageal mucosa from reflux of gastric juice (= GERD). Commonest in adults over 40. Symptoms: heartburn, dysphagia, regurgitation of sour gastric contents; rarely severe chest pain mistaken for heart disease.
Causes	Anything that decreases LES tone and/or raises abdominal pressure : alcohol and tobacco, obesity, CNS antidepressants, pregnancy, hiatal hernia, delayed gastric emptying, increased gastric volume. Often no cause found.
Complications	Ulceration · hematemesis and melena · stricture · Barrett esophagus .



Reflux: hyperemia, redness



Reflux histology: basal zone hyperplasia



A reflux (scattered eosinophils) · B eosinophilic (numerous)

4 GERD → Barrett → adenocarcinoma

THE BLOCK'S RECURRING PATTERN

Chronic injury → **metaplasia** → dysplasia → carcinoma. Here: long-standing GERD → Barrett (glandular metaplasia) → dysplasia → adenocarcinoma. You will meet it again as leukoplakia → SCC, H. pylori → intestinal metaplasia → gastric cancer, and adenoma → colorectal cancer.

Long-standing GERD

Barrett: glandular metaplasia

Dysplasia

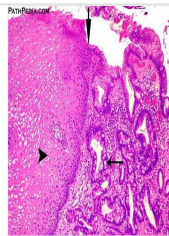
Adenocarcinoma

Barrett esophagus

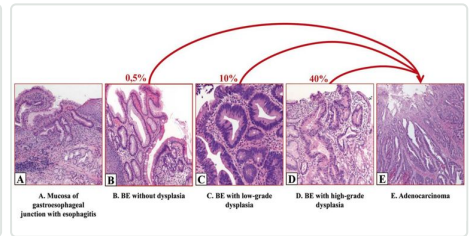
Acquired glandular metaplasia of the esophagus from reflux of gastroduodenal contents in long-standing GERD. Risk of progression to adenocarcinoma; **dysplasia is the intermediate stage** between metaplasia and malignancy.



Barrett esophagus: gross



Barrett: glandular mucosa



Progression to adenocarcinoma (risk % on slide)

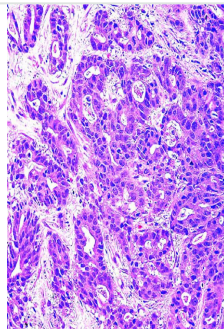
5 Esophageal tumours — adenocarcinoma vs SCC

Benign: epithelial (papilloma) or mesenchymal — **leiomyoma is the commonest benign tumour**. Almost all malignant tumours are epithelial: adenocarcinoma and SCC.

	Adenocarcinoma	Squamous cell carcinoma
Arises from	Barrett esophagus (most)	Squamous epithelium
Site	Distal third ; may invade gastric cardia	Middle third in half of cases
Trend	Incidence increasing worldwide	More common worldwide
Risk factors	Long-standing GERD → Barrett	Alcohol and tobacco · poverty · caustic injury · achalasia · Plummer-Vinson · very hot drinks · mediastinal radiation · nutritional deficiency · polycyclic hydrocarbons, nitrosamines, fungus-contaminated food · HPV in high-risk regions



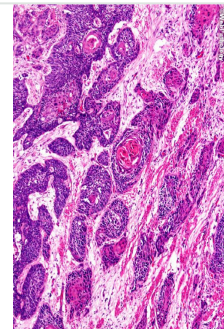
Adenocarcinoma: gross



Adenocarcinoma: glands



SCC: gross



SCC: histology

6 Clinical picture & spread

Presentation	Usually advanced at diagnosis. Dysphagia is the commonest symptom; odynophagia; weight loss and debilitation; hemorrhage and sepsis; iron deficiency symptoms; aspiration of food via a tracheoesophageal fistula.
Why it spreads	A rich submucosal lymphatic network promotes circumferential and longitudinal spread.
Tumour location	Lymph nodes involved
Upper third	Cervical LN
Middle third	Mediastinal, paratracheal, tracheobronchial LN
Lower third	Gastric and celiac LN

P Must-remember pearls

- **Achalasia is a triad:** increased LES tone, incomplete relaxation, aperistalsis. Chagas (*T. cruzi*) is the named secondary cause.
- **Varices:** cirrhosis and portal hypertension in >90%; rupture risk above 5 mm.
- **Mallory-Weiss is mucosal, Boerhaave is transmural** — same trigger; the transmural one causes mediastinitis and is usually fatal.
- **Adenocarcinoma = distal third, from Barrett; SCC = middle third, more common worldwide.** Dysplasia is the step between Barrett and adenocarcinoma.
- **Dysphagia** is the commonest symptom of esophageal carcinoma — and it usually presents late.

Q Test yourself

1. A 46-year-old man with long-standing heartburn has a biopsy showing glandular epithelium above the GE junction. Which malignancy is he at increased risk of, and where does it usually arise?

- A. Adenocarcinoma of the distal third
- B. SCC of the middle third
- C. Leiomyoma of the lower third
- D. SCC of the upper third
- E. Adenocarcinoma of the upper third

2. A patient from South America has dysphagia. Manometry shows raised LES tone, incomplete LES relaxation and aperistalsis. Which organism should be considered?

- A. *Candida albicans*
- B. Herpes simplex virus
- C. Cytomegalovirus
- D. *Trypanosoma cruzi*
- E. Human papillomavirus

3. Two patients vomit repeatedly after heavy drinking. One has hematemesis; the other has severe chest pain, tachypnea and shock. What distinguishes the second patient's lesion?

- A. It is longitudinal and near the GE junction
- B. It is caused by portal hypertension
- C. It is transmural rather than mucosal
- D. It arises in the proximal esophagus
- E. It has a different mechanism from the first

4. An immunosuppressed patient has odynophagia. Biopsy shows multinucleate squamous cells containing nuclear inclusions. The most likely cause is:

- A. *Candida*
- B. Eosinophilic esophagitis
- C. Reflux esophagitis
- D. Radiation
- E. Herpes simplex virus

5. A cirrhotic patient has hematemesis. Which statement about his esophageal varices is correct?

- A. They lie in the mucosa, not the submucosa
- B. They rupture more readily when larger than 5 mm
- C. They are usually due to schistosomiasis
- D. They form in the proximal esophagus
- E. The overlying mucosa is always ulcerated

ANSWERS — COVER UNTIL YOU'VE COMMITTED

- 1 • A** — Barrett esophagus (glandular metaplasia from chronic GERD) predisposes to adenocarcinoma, usually in the distal third.
- 2 • D** — Secondary achalasia in Chagas disease: *T. cruzi* destroys the myenteric plexus, producing the same triad as primary achalasia.
- 3 • C** — Boerhaave syndrome is a transmural rupture with severe mediastinitis; Mallory-Weiss is a mucosal tear. Similar mechanism, different depth.
- 4 • E** — Herpes esophagitis: shallow vesicles and ulcers with multinucleate squamous cells and herpesvirus nuclear inclusions. Candida gives gray-white pseudomembranes of hyphae.
- 5 • B** — Varices in the submucosal plexus of the distal esophagus (cirrhosis in >90%) may rupture when >5 mm; overlying mucosa is intact until rupture.

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam

Pathology GIT Block · MEDucation by Mattin Majid · summarised from LG2, Dr Jawher Tahir Omer, Hawler Medical University
Study alongside the lecture, not instead of it.